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Industry Run-Ups and Subsequent Crash Risk: Evidence from the German Equity Market

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List of Abbreviations

AI	Artificial intelligence.
DEM	ISO currency code for the Deutsche Mark.
EFE	Empirical Financial Economics (seminar title).
et al.	And others; used in citations for works with multiple authors.
EUR	Euro; ISO currency code.
FRED	Federal Reserve Economic Data.
GICS	Global Industry Classification Standard.
GSY	Greenwood, Shleifer, and You (2019); shorthand for the original study.
gvkey	Company identifier used in Compustat Global.
ID	Identifier; used in "student ID".
iid	Issue identifier used in Compustat Global.
n.d.	No date; used in reference entries without a publication date.
N/A	Not available or not applicable, depending on context.
OECD	Organisation for Economic Co-operation and Development.
pp	Percentage points.
RF	Risk-free rate; in this study, an approximate risk-free-rate proxy.
Std. dev.	Standard deviation.
U.S.	United States.
WRDS	Wharton Research Data Services.

List of Symbols

α	Intercept in the linear probability model.
β	Coefficient on the run-up indicator; measures its conditional association with crash probability.
γ	Coefficient on pre-event industry volatility.
$\varepsilon_{i,t}$	Regression error term for industry i in month t.
i	Index for a GICS sector/industry portfolio.
t	Run-up or event month; in test statistics, t denotes the t-statistic.
$t + 24$	Subsequent 24-month outcome window following the run-up month.
$t = 0$	Normalized event month in Figure 1, where the wealth indices are set to one.
Crash indicator	Equals one if industry i experiences a raw drawdown of at least 40% within the following 24 months; zero if the complete window contains no such crash; missing if the window is incomplete.
$\mathbf{Runup}_{i,t}$	Run-up indicator at the specified threshold and under the definition used in the relevant regression.
$\mathbf{Volatility}_{i,t}$	Annualized volatility based on preceding daily industry returns.
N	Number of observations or qualifying episodes, as applicable.
p	p-value associated with a statistical test.
z	Test statistic for the two-proportion z-test.
χ^2	Pearson chi-square test statistic.
R^2	Coefficient of determination for a regression.
$\sqrt{252}$	Annualization factor used for daily volatility, assuming 252 trading days per year.

1 Introduction

1.1 Motivation and Research Question

Whether large price increases signal speculative bubbles remains a central question in finance. Fama (2014) emphasises the difficulty of identifying mispricing from observed prices, since strong appreciation may reflect changing expectations rather than irrational behaviour. Consequently, a subsequent decline cannot be inferred from a price increase alone.

Goetzmann (2016) highlights the danger of focusing on historical booms that ended in crashes while overlooking those that did not. Greenwood et al. (2019) address this distinction empirically, finding that large industry run-ups do not necessarily predict unusually low average returns but are associated with a substantially higher probability of a subsequent crash. This paper examines whether the same relationship holds in Germany. The central research question is whether a 100% industry price increase over the preceding 24 months predicts a higher probability of a 40% drawdown within the following two years, including after controlling for pre-event volatility.

1.2 Empirical Scope and Contribution

The study applies the empirical framework of Greenwood et al. (2019) to German equities using Compustat Global Daily data covering an industry-month panel from January 1986 to September 2026. Securities are aggregated into value-weighted GICS sector portfolios, with Deutsche Mark prices converted into euros. Run-up thresholds of 50%, 75%, 100%, 125%, and 150% are examined. Subsequent performance is evaluated through forward returns, crash probabilities, and maximum drawdowns. A German three-month interbank interest rate serves as an approximate risk-free-rate benchmark. The analysis provides country-specific evidence on whether the distinction between average-return predictability and crash risk documented by Greenwood et al. (2019) also appears in Germany.

1.3 Overview of Findings and Roadmap

At the strict 100% run-up threshold, three of four German industry episodes with complete 24-month follow-up experience a raw drawdown of at least 40%, corresponding to a crash frequency of 75.0%. The comparable frequency among eligible non-qualifying sector-months is 11.45%, while the unconditional frequency is 12.06%. The difference is

economically large, but the conditional estimate relies on only four episodes and should be interpreted cautiously.

Mean excess returns relative to the risk-free proxy following a 100% run-up are 15.0% over 12 months and 35.1% over 24 months; neither is statistically distinguishable from zero at conventional levels. In the broader sector-month regressions, the run-up indicators remain positively associated with subsequent crash probability after controlling for pre-event volatility. These results distinguish average-return performance from downside risk, but they do not establish a causal relationship. The paper develops the theoretical framework, describes the German data and empirical design, presents the results, and discusses their limitations.

2 Theoretical Background and Hypotheses Development

2.1 Market Efficiency and Limits of Arbitrage

The efficient market hypothesis provides a starting point for evaluating whether sharp price increases predict subsequent losses. Fama (1970) argues that security prices incorporate available information, while Fama (2014) discusses the role of market efficiency and asset-pricing models in explaining observed returns. From this perspective, a substantial price increase does not by itself demonstrate that an industry is overvalued, since changing expectations about fundamentals may justify higher prices.

Nevertheless, mispricing may persist when investors face constraints on correcting it. Shleifer and Vishny (1997) show that arbitrageurs face funding and investment risks that can limit their ability to exploit apparent pricing errors. Consequently, even if a run-up reflects excessive optimism, its duration and eventual correction need not be predictable with precision. This distinction motivates testing crash probabilities separately from average forward returns.

2.2 Investor Extrapolation and Behavioral Explanations of Bubbles

Behavioural explanations provide a possible mechanism linking strong past performance to future crash risk. Barberis et al. (1998) develop a model in which investors' belief formation can generate overreaction to sequences of positive or negative information. Applied to industry portfolios, extrapolative expectations may lead investors to interpret recent price appreciation as evidence that favourable performance will continue, potentially contributing to excessive valuations.

Greenwood and Shleifer (2014) find that survey-based return expectations rise with past market returns, consistent with extrapolative beliefs. Lehnert (2020) finds that U.S. industry run-ups ending in crashes are preceded by a shift from euphoria towards fear, whereas non-crash run-ups show no comparable pattern. This provides a possible explanation for divergent price paths, although the German analysis does not directly test investor sentiment. Greenwood et al. (2019) likewise show that run-ups need not predict unusually low average returns, even when crash probabilities rise.

2.3 Derivation of Empirical Hypotheses

These arguments motivate three hypotheses for the German equity market.

H1 — Average forward returns: Mean 12-month and 24-month returns in excess of the risk-free-rate proxy are not statistically different from zero following large industry run-ups.

H2 — Conditional crash probability: Industries experiencing a 100% price increase over the preceding 24 months have a higher probability of a subsequent 40% drawdown within two years than non-run-up observations.

H3 — Incremental predictive association: The positive association between a qualifying run-up and subsequent crash probability remains after controlling for pre-event industry volatility.

The first hypothesis addresses average performance, whereas the second concerns the probability of an extreme downside event. The third examines whether the relationship remains after accounting for differences in prior volatility. Together, these tests distinguish evidence of negative average-return predictability from evidence of elevated crash risk, following the central empirical distinction in Greenwood et al. (2019).

3 Data and German Industry Portfolio Construction

3.1 German Equity Sample Selection (Compustat Global Daily)

The analysis uses security-level observations from Compustat Global Daily accessed through WRDS. The initial sample contains 8,520,923 observations of German common equity with positive closing prices. Excluding 142,092 observations without a valid GICS sector classification leaves 8,378,831 observations. The subsequent currency restriction retains 8,096,137 EUR- and DEM-denominated observations. Securities are identified by the combination of gvkey and iid, allowing prices to be ordered consistently within each security. Table 1 summarises the sample construction and the reduction from daily observations to the final industry-month panel.

Table 1

Sample Construction and Data Coverage

Sample selection step	Observations	Description
German common equity before sector exclusion	8,520,923	German common equity with positive closing price
Excluded observations	−142,092	Missing or blank GICS sector
Retained German equity sample	8,378,831	Sector classification available
Excluded observations	−282,694	Currency other than EUR or DEM
EUR and DEM observations	8,096,137	Other currencies excluded
EUR observations	7,465,520	Euro-denominated
DEM observations	630,617	Converted at EUR 1 = DEM 1.95583
Valid adjusted total-return prices	8,096,137	Daily security observations
Valid daily returns before cleaning	8,093,558	Returns requiring a valid lagged price
Observations flagged by primary screen	4,176	Flagged adjusted prices and returns are set to missing; returns immediately following flagged prices are also set to missing
Month-end security observations	391,709	Monthly security panel
Unique securities	2,579	Identified by gvkey and iid
Final industry-month observations	5,266	Value-weighted portfolio panel
GICS sectors	11	Final industry classification

Note: The primary-screen categories overlap; 3,068 observations belong to reversal pairs and 1,788 meet the single-day threshold of at least 200%. Their union is 4,176 observations. The expanded screen flags 8,183 observations in total, including 4,007

flagged only by the expanded screen. Records are retained for auditing, although flagged adjusted prices and returns are set to missing.

3.2 Currency Harmonization (EUR/DEM Fixed Conversion)

The sample spans the transition from the Deutsche Mark to the euro. To ensure monetary consistency, DEM-denominated prices are converted at the fixed rate of EUR 1 = DEM 1.95583, while EUR-denominated prices remain unchanged (European Commission, n.d.). Adjusted total-return prices incorporate the reported closing price, the adjustment factor, and the total-return factor. Daily and monthly returns are calculated from these adjusted prices. The currency diagnostic records 500 transitions within security histories, which are treated as changes in reported currency rather than investment returns.

3.3 Rule-Based Daily Return Data Cleaning

Daily returns are screened using a primary rule: single-day returns of at least 200% are flagged, as are both observations in a two-day reversal pair when either return exceeds 100% and the compounded two-day return is below 20%. Flagged returns, adjusted total-return prices, and returns immediately following a flagged price are set to missing; dated records are retained.

The primary screen flags 4,176 observations. The expanded screen flags 8,183 observations in total, including 4,007 flagged only by the expanded screen. These screens target potential errors but cannot establish that every extreme return is erroneous (Ince & Porter, 2006). Separately, for volatility estimation only, eligible daily security returns are winsorised cross-sectionally across securities on each trading date at the nearest-rank 1st and 99th percentiles. The capped values are used only to construct daily industry returns for Step 9; this procedure does not overwrite the stored daily returns or adjusted prices and does not alter the monthly returns used for run-up identification or crash outcomes.

3.4 Construction of Value-Weighted Industry Portfolios and Market Benchmark

Monthly security returns are aggregated into value-weighted portfolios across 11 GICS sectors, using lagged market capitalisation. The German market benchmark is

constructed from the same eligible securities. Monthly net-of-market returns are calculated as the difference between the industry and market returns in the same month. For run-up identification, the 24-month signal is calculated as the compounded raw industry return minus the compounded market return over the same horizon. This distinction preserves the difference between monthly active returns and the cumulative two-year return criterion. The risk-free benchmark is an approximate German three-month interbank rate obtained from the OECD series distributed through FRED (OECD, 2026). The August 2026 rate is carried forward to September because the source series ends one month before the final industry panel month.

Table 2
Descriptive Statistics of Monthly Industry Returns

All statistics are expressed in percent per month.

Variable	N	Mean	Std. dev.	Min.	1st pct.	Median	99th pct.	Max.
Industry return	5,266	0.73	6.46	−36.40	−18.28	0.92	16.99	44.25
German market return	5,266	0.65	5.13	−22.85	−14.89	1.12	12.92	16.22
Net-of-market industry return	5,266	0.08	4.77	−32.67	−12.67	0.01	14.14	45.83
Industry excess return over RF proxy	5,266	0.49	6.48	−36.68	−18.64	0.69	16.72	43.98

Note: Net-of-market returns are calculated relative to the German market benchmark. Industry excess returns subtract the monthly risk-free-rate proxy. The proxy is derived from the German three-month interbank rate. The September 2026 rate is carried forward from August because the source series ends one month before the industry panel.

The resulting panel contains 5,266 industry-month observations across 11 sectors from January 1986 to September 2026. Table 2 shows that mean monthly industry returns are 0.73%, compared with 0.65% for the market and 0.49% for industry returns in excess of the risk-free-rate proxy. The proxy is derived from the annual percentage interbank rate by dividing by 1,200; the August 2026 rate is carried forward to September. As security-level observations extend only through 14 September 2026, the final industry-month observation may be partial. The substantial dispersion and negative tail observations motivate the subsequent analysis of run-ups and crash risk.

4 Empirical Methodology

4.1 Multi-Month Price Run-Up Identification Thresholds (50% to 150%)

Following Greenwood, Shleifer, and You (2019), a qualifying run-up must satisfy three cumulative-return conditions: the raw industry return over the preceding 24 months must be at least the specified threshold; the compounded raw industry return minus the compounded market return over the same period must meet that threshold; and the raw industry return over the preceding 60 months must be at least 50%. Thresholds of 50%, 75%, 100%, 125%, and 150% are examined. The sector must also contain at least ten eligible firms at the event date.

The event month is the first month in which these criteria are met. To avoid repeatedly counting a sustained run-up as separate episodes, another event in the same sector is not selected until at least 24 months after the previously selected event. This differs from suppressing a qualifying month merely because any qualifying month appeared in the previous 24 months. The risk-free-rate benchmark does not determine event eligibility. Forward outcome windows begin after the selected event month, and their completeness is assessed separately.

4.2 Definition of Subsequent Crashes and Booms (40% Drawdown Rule)

A crash is defined as a maximum peak-to-trough raw industry drawdown of at least 40% between the selected event month and the end of the subsequent 24-month window. The event month provides the initial wealth reference, and the crash indicator is observed only when a complete 24-month window is available. An event with incomplete follow-up is not classified as a non-crash.

A boom is defined as a compounded 24-month raw industry return exceeding the corresponding risk-free-rate proxy by at least 40 percentage points, using the study's return-difference measure. The crash and boom indicators capture different dimensions of performance: an industry can experience a severe interim drawdown and still finish the horizon with a positive return. Both outcomes are therefore reported alongside average forward returns.

4.3 Conditional Crash Probability and Regression Specifications

To examine whether strong prior industry performance predicts subsequent crash risk, a linear probability model is estimated at the industry-month level:

$$Crash_{i,t}^{(24)} = \alpha + \beta Runup_{i,t} + \gamma Volatility_{i,t} + \varepsilon_{i,t}$$

Here, $Crash_{i,t}^{(24)}$ equals one if industry i experiences a raw 40% drawdown during the 24 months following event month t, zero if the complete window contains no such crash, and missing if the window is incomplete. $Runup_{i,t}$ is the separate raw-return or net-of-market indicator used in each regression. $Volatility_{i,t}$ is annualised volatility from the preceding daily industry returns. Standard errors are clustered by calendar year.

5 Empirical Results

5.1 Descriptive Analysis of Historical German Run-Up Episodes

Table 3 reports the four selected German industry episodes meeting the strict 100% run-up criteria and having complete 24-month follow-up.

Table 3

Historical German Industry Episodes Meeting the Strict 100% Run-Up Criteria

Industry name	Number of firms	Price run-up first observed	12-month raw return (percent)	24-month raw return (percent)	12-month net of risk-free return (percent)	24-month net of risk-free return (percent)	12-month net of market return (percent)	24-month net of market return (percent)	24-month maximal drawdown (percent)	Months to price peak	Return to peak (percent)
Information Technology	11	1995-07	-10.1	69.6	-13.9	62.5	-19.1	-8	-17.8	24	69.6
Information Technology	15	1998-02	11.9	149.5	8.4	142.8	4.1	95.7	-42.4	24	149.5
Communication Services	10	1999-01	80	-15	77	-22.7	57.8	-27.1	-68.1	13	126.6
Real Estate	16	1999-07	-7.6	-33.3	-11.4	-42	-33.6	-36	-44.6	4	16
Crash mean			28.1	33.7	24.7	26	9.4	10.8	-51.7	13.7	97.4
All strict run-up mean			18.5	42.7	15	35.1	2.3	6.1	-43.2	16.2	90.4

Note: The table contains four selected strict 100% run-up episodes, all with complete 24-month outcomes. Three episodes meet the raw 40% crash definition. The first qualifying month identifies the event; “months to peak” refers to the subsequent peak within the event window. RF denotes the approximate risk-free-rate proxy.

The episodes concern Information Technology in July 1995 and February 1998, Communication Services in January 1999, and Real Estate in July 1999. Three of the four experience a subsequent raw drawdown of at least 40%.

The outcomes demonstrate substantial variation after similar prior appreciation. The average maximum drawdown across all four episodes is −43.2%, compared with −51.7% among the three crash episodes. Mean 24-month raw returns are 42.7% across all four episodes and 33.7% among crash episodes, illustrating that a severe interim drawdown does not necessarily imply a negative final holding-period return. The table is descriptive: with only four episodes, it cannot establish a stable pattern for the wider German equity market.

5.2 Threshold-Level Crash Probabilities and Forward Performance

Table 4 reports subsequent returns, crash frequencies, drawdown severity, and boom frequencies for run-up thresholds between 50% and 150%.

Table 4

Threshold-Level Forward Performance, Crashes and Booms

Run-up threshold	Run-up episodes identified	Episodes with complete 24-month follow-up	12-month net of risk-free return (%)	24-month net of risk-free return (%)	12-month net of market return (%)	24-month net of market return (%)	Standard deviation of 24-month net of risk-free return (%)	Skewness of 24-month net of risk-free return	Kurtosis of 24-month net of risk-free return	Crashes (%)	Average drawdown among crashes (%)	Booms (%)	Mean 24-month net of risk-free return among booms (%)
Unconditional	N/A	N/A	6.4	13.3	0.2	0.3	37.4	1.035	9.106	12.1	-52.6	20.4	66
[t-statistic]			[2.52]	[3.20]	[0.46]	[0.34]							
50%	15	13	12.8	1.6	7.1	-8.9	47	0.101	2.407	46.2	-57.9	23.1	63.7
[t-statistic]			[1.07]	[0.11]	[0.94]	[-1.32]							
75%	5	5	3.9	2.5	-2.4	-6.2	70.3	0.527	2.044	80	-58.4	40	73.1
[t-statistic]			[0.19]	[0.07]	[-0.21]	[-0.31]							
100%	4	4	15	35.1	2.3	6.1	84.9	0.67	1.132	75	-51.7	50	102.6
[t-statistic]			[1.06]	[0.66]	[0.26]	[0.17]							
125%	4	4	4.3	20.6	0.5	4.2	71.3	0.178	-0.349	75	-52.4	50	79.2
[t-statistic]			[0.50]	[0.46]	[0.08]	[0.15]							
150%	2	2	-15.7	21.1	-20.5	20.5				100	-56.7	50	78.4
[t-statistic]													

Note: Returns and drawdowns are expressed in percentages; skewness and kurtosis are unit-free. The unconditional row uses 3,881 eligible sector-months with complete 24-month outcomes, at least ten firms, and available 24- and 60-month return histories. Step 8 reports 4,978 complete windows before these additional eligibility restrictions. Bracketed values are year-clustered t-statistics for mean returns. Estimates at higher thresholds are descriptive and sensitive to small sample sizes.

Complete-outcome samples contain 13, 5, 4, 4 and 2 episodes at the 50%, 75%, 100%, 125% and 150% thresholds. The corresponding crash frequencies are 46.2%, 80.0%, 75.0%, 75.0% and 100.0%. The relationship is not monotonic, and the high-threshold estimates are sensitive to individual observations.

At the main 100% threshold, mean returns in excess of the risk-free proxy are 15.0% over 12 months and 35.1% over 24 months. The displayed year-clustered t-statistics are 1.06 and 0.66, respectively. However, these estimates are based on only four episodes spanning three calendar years, making formal inference highly uncertain. The results do not establish that mean excess returns differ from zero. Mean market-relative returns are also not statistically significant in the reported results.

Three of four strict 100% episodes experience a qualifying crash, with an average maximum drawdown of −51.7%. At the same threshold, 50% of episodes meet the boom definition. These results illustrate that severe interim losses and substantial cumulative gains can coexist. However, the 100% estimate rests on four observations and the 150% estimate on two; neither should be treated as a precise estimate of German crash probability. The unconditional row provides the benchmark for eligible sector-months with complete 24-month outcomes.

Table 5 formally compares the crash probability in the strict 100% run-up sample with that in the disjoint non-run-up sample.

Table 5

Formal Tests of Conditional Crash Probability

Sample	Observations	Crashes	Crash probability (%)	Difference vs non-qualifying (pp)	Two-proportion z	z-test p-value	Pearson chi-square	chi-square p-value	Fisher exact p-value
All eligible sector-months (context)	3881	468	12.06						
Selected strict 100% events	4	3	75	63.55	3.98	0.000069	15.838	0.000069	0.00555
Qualifying signals suppressed by spacing rule	34	25	73.53						
Eligible non-qualifying sector-months	3843	440	11.45						

Note: The selected-event group and the eligible non-qualifying group are used for the formal comparison. The 34 suppressed observations are qualifying sector-month signals excluded from both comparison groups because they overlap or recur within the event-spacing rule; they are not 34 independent additional events. All observations require complete 24-month outcomes. The baseline tests do not adjust for serial dependence across sector-month outcomes.

The four selected strict 100% run-up episodes have three crashes, yielding a conditional crash frequency of 75.0%. Among eligible non-qualifying sector-months, 440 of 3,843 observations experience a crash, corresponding to 11.45%. The difference is 63.55 percentage points. The asymptotic two-

proportion and Pearson chi-square tests both yield \(p=0.000069\), whereas Fisher's exact test yields \(p=0.00555\). Given that the selected group contains only four events, Fisher's exact result provides a useful reference for interpreting the difference. All tests should be treated cautiously because sector-month outcomes can overlap and may be serially dependent.

A further 34 sector-month signals meet the 100% qualifying criteria but are suppressed by the event-spacing rule; 25 of these signals are followed by a crash outcome. They are reported separately because overlapping signal months are not independent selected episodes. Although the observed difference between selected events and non-qualifying months is large, the selected group contains only four events, and the baseline tests do not adjust for serial dependence. The results should therefore be interpreted as evidence of a strong association in this sample rather than a causal or universally applicable effect.

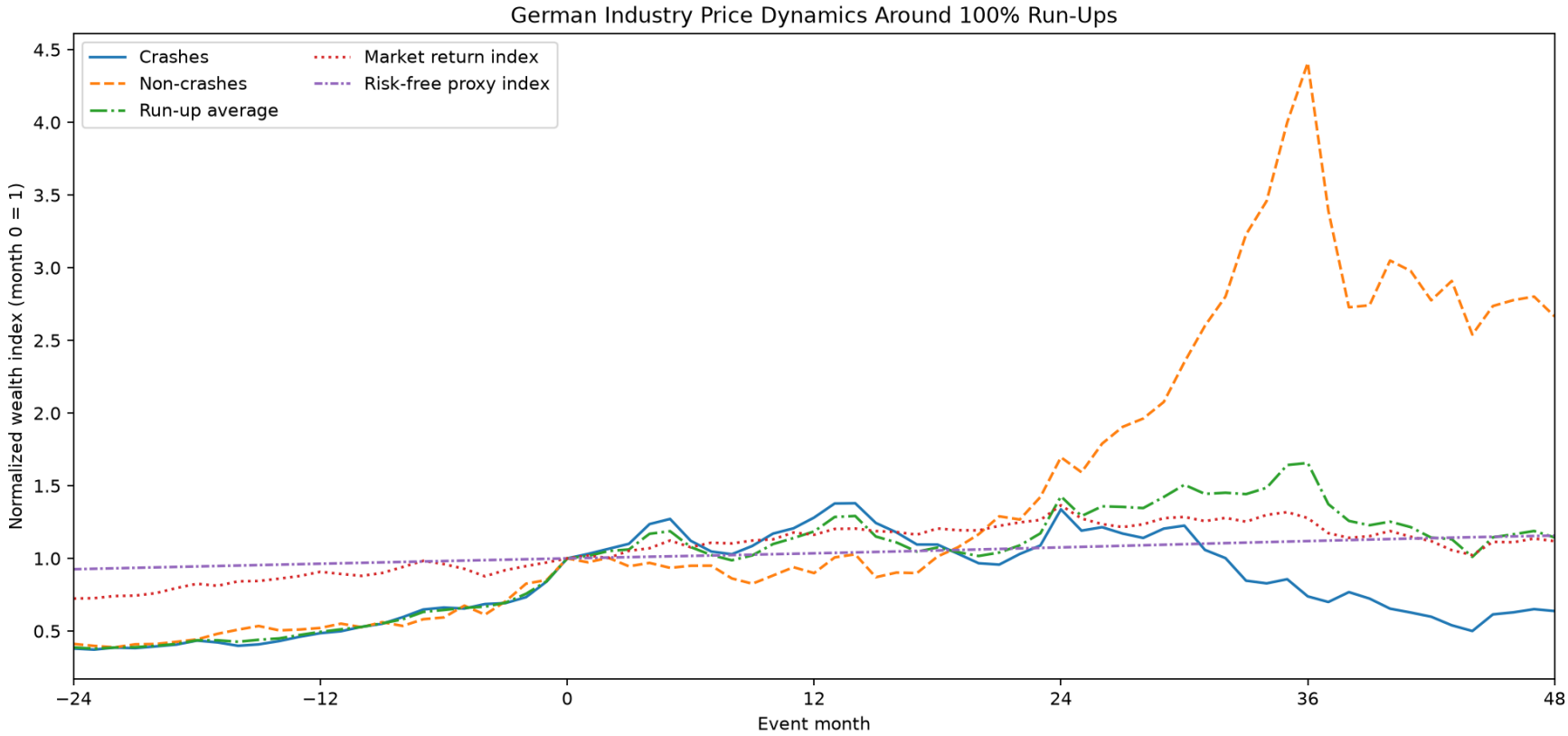
5.3 Event-Time Price Dynamics Around Run-Ups

Figure 1 traces industry wealth paths from 24 months before to 48 months after the selected 100% run-up month. All paths are normalised to one at event month zero. The figure compares the three episodes that subsequently experience a 40% raw drawdown with the single non-crash episode and includes the German market and risk-free proxy as benchmarks.

The figure illustrates that industries with similarly strong prior appreciation can subsequently follow different paths. The crash episodes experience severe interim declines, while the non-crash episode does not cross the 40% drawdown threshold. However, only four episodes enter the figure, and the number of available paths can vary across relative months when return observations are missing. The plot is therefore descriptive evidence rather than an independent statistical test or a reliable basis for forecasting the timing of a crash.

Figure 1

Event-Time Price Dynamics Around German Industry Run-Ups



Note: Normalised industry wealth indices from event month −24 to +48, with wealth equal to one at event month zero. Lines show crash episodes, the non-crash episode, the average run-up path, the German market benchmark, and the risk-free proxy. The strict 100% run-up sample contains four episodes with complete 24-month crash outcomes. The event month marks the selected run-up month, not necessarily the subsequent price peak.

5.4 Regression Evidence and Pre-Run-Up Volatility Controls

Table 6

Run-Up Indicators and Subsequent Crash Probability

Run-up measure	Threshold (%)	N (without volatility)	Run-up coefficient (without volatility)	p-value (without volatility)	N (with volatility)	Run-up coefficient (with volatility)	p-value (with volatility)	Volatility coefficient	Volatility p-value	R-squared (with volatility)	Calendar-year clusters (without volatility)	Calendar-year clusters (with volatility)
Raw return	50	4124	0.1572	0.0642	4124	0.1528	0.0616	0.8612	0.0371	0.0769	38	38
Raw return	75	4124	0.2446	0.0243	4124	0.2171	0.0314	0.7993	0.0541	0.0745	38	38
Raw return	100	4124	0.4281	0.0046	4124	0.3578	0.0081	0.7389	0.0708	0.0751	38	38
Raw return	125	4124	0.5340	0.0018	4124	0.4360	0.0050	0.7530	0.0712	0.0692	38	38
Raw return	150	4124	0.6955	<0.0001	4124	0.5735	<0.0001	0.7660	0.0717	0.0685	38	38
Net-of-market return	50	4124	0.3382	0.0301	4124	0.2819	0.0349	0.7664	0.057	0.0700	38	38
Net-of-market return	75	4124	0.6607	0.0001	4124	0.5558	0.0008	0.6919	0.0894	0.0836	38	38
Net-of-market return	100	4124	0.6265	0.0003	4124	0.5101	0.0018	0.7609	0.0725	0.0677	38	38
Net-of-market return	125	4124	0.6572	<0.0001	4124	0.5308	0.0004	0.7909	0.0657	0.0621	38	38

Net-of-market return	150	4124	0.7618	<0.0001	4124	0.6156	<0.0001	0.8076	0.0639	0.0593	38	38

Note: The dependent variable indicates a raw 40% drawdown within the following 24 months. Each regression separately thresholds the 24-month raw or net-of-market return. Unlike the strict event analysis in Tables 3–5, the regressions do not impose the joint raw-return and net-of-market criteria, the five-year raw-return requirement, or the event-spacing rule. The regression sample requires a complete outcome window, at least ten eligible firms, and the return history required to construct the relevant 24-month indicator. It does not require the additional 60-month history used for the unconditional sample in Tables 4–5. All specifications use 4,124 observations and 38 calendar-year clusters.

Table 6 reports linear probability models relating subsequent raw 40% drawdown crashes to separate raw-return and net-of-market run-up indicators at five thresholds. Specifications with volatility include annualised volatility calculated from the preceding daily industry returns, constructed after winsorising eligible security returns at the daily cross-sectional 1st and 99th percentiles. All specifications use 4,124 observations and standard errors clustered across 38 calendar years.

At the 100% threshold, the raw-return coefficient is 0.3578 ($p = 0.0081$), while the net-of-market coefficient is 0.5101 ($p = 0.0018$). These estimates correspond to conditional differences in crash probability of 35.78 and 51.01 percentage points, respectively. The controlled coefficients remain positive and statistically significant at the 5% level across all thresholds except the raw-return 50% specification ($p = 0.0616$). Volatility is statistically significant at the 5% level only in the raw-return 50% specification. The R-squared values range from 0.059 to 0.084. Overall, the regressions indicate a positive conditional association between run-ups and subsequent crash risk, without establishing causality.

6 Robustness, Discussion, and Limitations

6.1 Sensitivity Across Run-Up Thresholds and Specifications

The threshold analysis reveals elevated crash frequencies following large industry run-ups, but the estimates do not rise monotonically with the threshold. Among episodes with complete 24-month outcomes, crash frequencies are 46.2%, 80.0%, 75.0%, 75.0%, and 100.0% at thresholds of 50%, 75%, 100%, 125%, and 150%, respectively. The corresponding complete-event counts are 13, 5, 4, 4, and 2.

The 100% threshold is the principal specification because it balances an economically large prior return with the available German event sample. Three of its four episodes experience a crash, but the resulting 75.0% estimate is sensitive to individual outcomes. The 150% estimate is based on only two events and should not be interpreted as a precise estimate of crash probability at that threshold. The expanded screen flags additional potential data errors but is not accompanied by a full re-estimation of the outcome results; it is therefore a cleaning diagnostic rather than a full outcome robustness test.

6.2 Comparison of German Findings with GSY (2019) U.S. Results

Table 7 compares the German crash frequencies with those reported by Greenwood, Shleifer, and You (2019). At the 100% threshold, the German sample records a 75.0% crash frequency, compared with 53% in the original U.S. sample and 50% in the international sample. At the 50% threshold, the German frequency is 46.2%, compared with 20% in the U.S. sample and 36% internationally. The German estimates exceed both the U.S. and international values at the 50%, 75%, 100%, and 150% thresholds. At 125%, the German crash frequency of 75.0% is slightly below the U.S. estimate of 76%, but above the international estimate of 53%. The German denominators are particularly small at higher thresholds: four episodes at 100% and two at 150%.

Table 7

Subsequent Crash Probabilities Following Industry Run-Ups: Germany and GSY (2019)

Run-up threshold	Germany: episodes	Germany: crashes	Germany: crash probability (%)	GSY U.S.: episodes	GSY U.S.: crash probability (%)	GSY international: episodes	GSY international: crash probability (%)
50%	13	6	46.2	168	20	237	36
75%	5	4	80.0	77	36	153	42
100%	4	3	75.0	40	53	107	50
125%	4	3	75.0	21	76	74	53
150%	2	2	100.0	15	80	51	67

Note: German episodes are counted only when the full 24-month outcome is observed. GSY U.S. and international results are taken from Greenwood, Shleifer, and You (2019), Table 3, Panels A and B. The U.S. sample comprises industry portfolios; the international sample combines sector portfolios across countries. Differences in market coverage, sample period and the German sector definition make this a descriptive comparison rather than a formal cross-country test.

Forward-return results also differ. At the 100% threshold, German mean excess returns relative to the risk-free proxy are 15.0% over 12 months and 35.1% over 24 months, neither statistically distinguishable from zero. GSY report U.S. estimates of approximately 3% and −10%, respectively, and international estimates of approximately −2% and 2%. The German boom frequency is 50% at the 100% threshold, higher than the 18% U.S. and 26% international estimates reported by GSY. These differences should not be overinterpreted because the German sample is much smaller, uses broad GICS sectors, and covers a different period. The results indicate a similar distinction between average-return performance and crash incidence, but they do not establish that the German and original estimates are statistically different.

6.3 Implications for the Market Efficiency Debate

The German evidence reinforces the distinction between average-return predictability and the probability of extreme downside outcomes. At the strict 100% threshold, three of four selected episodes experience a raw drawdown of at least 40%, while mean excess returns relative to the risk-free proxy are not statistically distinguishable from zero. The crash frequency is 75.0%, compared with 11.45% among eligible non-qualifying sector-months, although the selected group is very small.

This pattern is consistent with the interpretation in Greenwood, Shleifer, and You (2019): large price run-ups may be associated with elevated crash risk without implying systematically negative average forward returns. The results do not establish that every large run-up is a speculative bubble, nor can they identify when a price peak will occur. They instead suggest that average returns and downside-event probabilities provide complementary information when evaluating unusually strong price appreciation.

6.4 Methodological and Data Limitations

The main limitation is the small number of qualifying German episodes. The strict 100% sample contains four episodes with complete 24-month follow-up, of which three experience a raw crash. The resulting 75.0% crash frequency is therefore sensitive to individual outcomes. The 150% estimate uses only two episodes and should be interpreted descriptively. The reported z, chi-square and Fisher tests are baseline comparisons and do not account for serial dependence across sector-month outcomes.

The analysis uses Compustat Global Daily data and value-weighted portfolios across 11 broad GICS sectors, whereas GSY use 48 Fama–French industry portfolios for the U.S. sample and international sector portfolios for their cross-country analysis. The shorter German sample and coarser industry classification reduce the opportunity to observe separate episodes and limit direct numerical comparability. The historical episode table now applies the same strict 100% run-up definition as the main event analysis rather than the former raw-only screen.

The primary screen flags 4,176 observations; the expanded screen flags 8,183, including 4,007 additional observations. These rules cannot establish that every flagged return is erroneous, and the expanded screen is not followed by a full outcome re-estimation. Winsorisation is restricted to the daily returns used for volatility, leaving monthly run-up and crash returns unchanged.

The risk-free benchmark is an approximate three-month interbank-rate proxy rather than a directly investable return. The August 2026 value is carried forward to September, while security-level observations end on 14 September 2026 and the final monthly observation may be incomplete. The analysis identifies conditional associations rather than causal effects, and its findings are limited to the observed German sample and the selected methodological definitions.

The regressions cluster standard errors by calendar year, allowing for dependence among observations within a year, but this approach does not fully address serial correlation within sectors across years. The regression p-values should therefore be interpreted with this limitation in mind.

7 Conclusion

7.1 Summary of Empirical Insights

This paper applies the empirical framework of Greenwood, Shleifer, and You (2019) to German equities, using value-weighted GICS sector portfolios and strict run-up criteria based on compounded raw and market-relative returns. At the 100% threshold, four distinct episodes have complete 24-month follow-up, and three experience a raw drawdown of at least 40%, giving a crash frequency of 75.0%. The corresponding frequency among eligible non-qualifying sector-months is 11.45%.

The forward-return evidence differs from the crash-frequency evidence. At the 100% threshold, mean excess returns relative to the risk-free proxy are 15.0% over 12 months and 35.1% over 24 months, neither statistically distinguishable from zero. The broader sector-month regressions also show a positive association between run-up indicators and subsequent crash probability after controlling for volatility, although the raw-return 50% coefficient is not significant at the 5% level in the controlled specification.

These results are consistent with the distinction between average-return predictability and downside-event risk emphasised by GSY. However, the small number of selected episodes, differences in industry classification and sample coverage, and the observational design prevent strong generalisations or causal conclusions. The German evidence should be interpreted as country-specific evidence from the available sector-month sample.

7.2 Concluding Remarks and Outlook

The results suggest that unusually strong industry price appreciation in Germany can be associated with elevated subsequent crash risk, even when average forward returns are not significantly different from zero. The strict 100% threshold has a crash frequency of 75.0%, but this estimate is based on only four complete episodes and is sensitive to individual outcomes. It should not be interpreted as a stable probability for future German industry run-ups.

The findings therefore qualify rather than reject the market-efficiency perspective: a large price increase alone does not establish mispricing, guarantee a crash, or reveal the timing of a price peak. Future research could extend the sample across countries, use

more disaggregated industry portfolios, and examine how alternative data-cleaning rules influence the identification of extreme events. Overall, the study highlights the value of evaluating average forward returns alongside the frequency and severity of downside events, while recognising the limitations imposed by the available German data.

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9 Appendix

9.1 Appendix A: Variable Definitions and Cleaning Rules

Table A.1 summarises the principal variables, their construction, and their role in the empirical analysis.

Table A 1

Variable Definitions and Construction

Variable	Definition and construction
tr_price_eur	Adjusted total-return price in euros, calculated from closing price, the currency-conversion factor, adjustment factor and total-return factor. Flagged adjusted prices are set to missing under the selected cleaning rule.
ret_daily	Daily return calculated from consecutive adjusted total-return prices minus one. Returns are set to missing when flagged by the primary screening rule or when they immediately follow a flagged adjusted price.
ret_monthly	Monthly return calculated from consecutive month-end adjusted total-return prices. It is missing when the preceding observed month is not the immediately preceding calendar month, or when the relevant adjusted prices are unavailable.
mktcap_lag	Previous dated security market capitalisation, calculated from the euro-converted closing price and shares outstanding. The lag is determined before missing daily returns are excluded.
industry_return	Monthly sector return, weighted by lagged market capitalisation.
market_return	Monthly German market benchmark return, calculated from the same eligible securities and lagged market-capitalisation weights.
net_market_return	Monthly industry return minus the German market return.
rf_monthly	Approximate monthly risk-free-rate proxy derived from the annual German three-month interbank rate, divided by 1,200.

Variable	Definition and construction
industry_excess_rf_return	Monthly industry return minus the monthly risk-free-rate proxy.
runup_raw24	Compounded raw industry return over the 24 calendar months ending in the event month.
runup_net24	Compounded raw industry return minus the compounded market return over the same 24-month window.
runup_raw60	Compounded raw industry return over the 60 calendar months ending in the event month.
qualify50–qualify150	Indicators for sector-month observations meeting the joint raw, market-relative, five-year, and minimum-firm criteria at each threshold, before event-spacing suppression.
runup50–runup150	Indicators for the selected non-overlapping episodes, obtained by selecting the first qualifying month and requiring at least 24 months between selected events in the same sector.
fwd12_raw, fwd24_raw	Compounded raw industry returns over the following 12 and 24 months.
fwd12_excess_rf, fwd24_excess_rf	Forward raw returns minus the corresponding compounded risk-free-rate benchmark.
fwd12_net, fwd24_net	Forward compounded industry returns minus the corresponding compounded market returns.
maxdd_raw24	Maximum peak-to-trough drawdown in the raw industry wealth path from the event month through the subsequent 24 months.
crash_raw40	Indicator equal to one if the maximum raw drawdown reaches at least 40% during the 24-month outcome window; defined only if the window is complete.
crash_net40	Indicator equal to one if the maximum market-relative drawdown reaches at least 40% during the 24-month outcome window; defined only if the window is complete.
volatility_annual	Annualised sample standard deviation of daily value-weighted sector returns over the preceding 12 months, multiplied by $\sqrt{252}$. Daily security returns are winsorised by trading date at the 1st and 99th percentiles before aggregation.
complete12, complete24	Indicators for whether the sector has all required calendar-month observations in the subsequent 12- or 24-month outcome window.

Daily-return cleaning

The primary screen flags returns of at least 200% and two-day reversal pairs meeting the specified 100% return and 20% cumulative-return criteria. Flagged returns, adjusted prices and immediately subsequent returns are set to missing; dated records are retained. The expanded screen flags 8,183 observations in total, including 4,007 flagged only by the expanded screen. Separately, eligible daily returns are winsorised by trading date at the 1st and 99th percentiles for volatility estimation.

Run-up identification.

A qualifying sector-month must satisfy the raw and market-relative 24-month return thresholds, a minimum 50% compounded raw return over 60 months, and a minimum of ten eligible firms. The selected event is the first qualifying month; another event in the same sector is not selected until at least 24 months later. Forward outcomes begin after the selected event month. Complete 24-month follow-up is required to classify a crash.

Risk-free-rate proxy.

The monthly benchmark is derived from the German three-month interbank interest-rate series distributed through FRED. It is an approximate benchmark rather than a directly investable return. The September 2026 rate is carried forward from August because the source series ends one month before the industry panel.

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